

Addressing Digital Divides in Rural Appalachia with Digital Literacy Education

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ABSTRACT

To highlight and contribute to the resolution of digital divides in rural Appalachia, I propose a series of qualitative studies taking place at public libraries in a rural county in West Virginia. These studies aim to identify and characterize the presence of intra-rural digital divides, which are disparities present between areas of similar rurality. I also evaluate digital storytelling as a culturally-resonant method of digital literacy education for Appalachian residents. In preliminary work towards these goals, I have developed a theoretical frame for this work, and conducted one initial study evaluating digital storytelling for digital literacy through a web-based storytelling system. In future work I intend to conduct, I will further evaluate digital storytelling's effectiveness for this audience through a new Microsoft Word plugin. I will also compare study sites within one rural county to identify and characterize intra-rural digital divides by highlighting qualitative differences between them in terms of digital access, digital literacy, and computer self-efficacy.

KEYWORDS

rural HCI, digital divide, digital literacy, digital storytelling

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1 PROPOSED RESEARCH

1.1 Research Problem and Objectives

Despite the ubiquity of computing and internet technology in the lives of many Americans, the digital divide still affects many members of rural communities, with computing and internet technology being either unavailable or outdated and expensive [17, 23]. This issue is particularly rampant in Appalachia [16, 19] – a geographical designation including areas in 13 states across the Eastern United States with a significant rural population [21]. Appalachia has been well-documented as a severely impacted region of the rural-urban digital divide in the United States [12, 16, 18]. While the digital divide is typically discussed in a rural-urban context, there is also

evidence to suggest that there is an intra-rural digital divide between rural communities with different characteristics [15, 25]. For example, 25.3% of rural Appalachian households did not have access to broadband internet in 2015-2019, compared to 20.5% of other rural households in the United States [21].

My proposed research qualitatively investigates the intra-rural digital divide in the US, addressing a gap in the existing field of literature on the intra-rural digital divide, which has so far been investigated quantitatively in international contexts [8, 15, 25]. I also intend to investigate mitigation strategies for these intra-rural digital divides by identifying effective and culturally-resonant means of digital literacy education for rural Appalachians.

Appalachia has been targeted by recent governmental efforts to extend broadband connectivity across disconnected regions based on its particularly low rates of broadband accessibility [12]. This makes digital literacy skills development increasingly important, as residents will need to have these skills to effectively utilize newly available internet technologies. To address digital inequality impacting residents of rural Appalachia, researchers have previously sought to increase digital literacy [20], but the unique characteristics of the region make specific, cultural considerations necessary for any interventions taking place there [4]. Prior work in rural HCI has also emphasized the importance of considering local culture when designing technology for rural users [1, 3, 5, 9–11, 22]. However, little prior work has been conducted on the design and deployment of region-specific digital literacy education in Appalachia. To combat digital inequality, my work aims to identify culturally appropriate and effective opportunities for digital literacy skills development for Appalachian residents.

1.2 Research Questions

- (1) How do differences in foundational values impact the researcher/participant relationship and how can this impact be mediated ethically?
- (2) How does culturally-tailored digital literacy education affect rural residents' technology self-efficacy?
- (3) How does poor infrastructure interact with self-efficacy to affect the development of digital literacy?
- (4) What disparities are present in computer and internet access, use, skills, and perception between different communities within the same rural county?
- (5) How do residents' perceptions of surrounding towns' resources and infrastructure impact their own technology self-efficacy?

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2 PRELIMINARY WORK

2.1 Designing for Shared Values: A Theoretical Framework for Design Research

In this work, I discussed provocations and directions for an approach to Participatory and Value Sensitive Design that recognizes the challenges of researching with participants whose core values conflict with those of the researchers. The approach I developed in this work centers around establishing a shared subset of values between participants and researchers that can guide design.

2.1.1 Outcome. In this work I made a conceptual and a methodological contribution. The conceptual contribution is a series of provocations to guide a conversation in the HCI and CSCW communities around ethics, power, and responsibility in design research. The methodological contribution of this work is a process based on these provocations called Designing for Shared Values (DSV). The DSV approach is intended for scenarios in which conflict between researchers and participants is unproductive and could lead to a rift in the research relationship. It centers design research around a commonly held set of values, building upon prior approaches for values negotiation which either support co-existence of conflicting values or establish consensus on conflicting values through reasoned debate [7, 14]. DSV also builds off of prior work in its focus on reflexivity as it necessitates researchers to identify their own values and their relationship to the research project, setting, and community [2, 13, 24]. Through this approach, we posited that by designing technologies grounded in values that researchers and participants can support, we can highlight our similarities to lay important groundwork for future productive conflict resolution to take place.

2.2 Evaluating Digital Storytelling as an Approach for Digital Literacy in Rural Appalachia

In this work, we built and deployed a digital storytelling system to teach digital literacy skills to residents of a rural Appalachian community. This work provided an initial investigation into the effectiveness of digital storytelling for digital skills development for an Appalachian audience.

2.2.1 Findings. Through our observations of participants' use of the system, we identified four key themes. First, we found that digital storytelling resonated with our participants and their Appalachian culture and heritage. Second, we found that engaging in digital storytelling through our system effectively taught computer skills to our participants and facilitated an opportunity for them to practice existing skills. Third, we found that participants experienced challenges when using technology due to low self-efficacy, lack of technical experience, mismatches between system design and participant use, and poor infrastructural conditions. Finally, we found that slow and unreliable computing infrastructure at our study site significantly and negatively impacted our participants' experience using computers. We observed that these poor infrastructural conditions went beyond a degraded user experience as compared to the experience of using reliable, high-speed technology, but transformed the user experience. This transformed experience

seemed to reinforce participants' low self-efficacy, creating an experience of use that directly conflicted with participants' ability to learn computer skills and develop digital literacy.

3 FUTURE WORK

Building upon this preliminary work, I will conduct two additional studies to achieve goals of investigating opportunities and challenges to digital literacy education in Appalachia, and evaluation of the impacts of observed intra-rural digital divides in the region. I will conduct these studies at six public libraries across one county in rural West Virginia.

3.1 Study 1 - Effective Digital Literacy Education Evaluation

In the first of these two studies, I will evaluate a Microsoft Word plugin I designed and developed in collaboration with my lab. This system takes stories that were recorded through the previous study I conducted, and transcribes them into editable files in Microsoft Word. Participants will learn how to edit these files for accuracy by listening back to the audio files that prior participants recorded, and making necessary changes to the Microsoft Word document. I anticipate that this study will reinforce findings from preliminary work, regarding the appropriateness of digital storytelling and the barriers that poor internet infrastructure imposes on learning. Because the digital literacy skills acquired through the Microsoft Word system are more specific and in depth, I expect that participants will be more aware of their skills acquisition than in the case of the storytelling website.

I will hold classes to teach community members some foundational principles of the technology they will be using, and instructions on how to use the storytelling site and Microsoft Word plugin. The observational data collected in these classes will be supported by additional observations and interviews of community members who use the plugin outside of the class setting. Additionally, during my visits to this study site to conduct these classes, I will provide general technology support in the form of a technology help desk where I will learn more about community members' desires and challenges with technology. This approach will help to fill in gaps in knowledge about what the community wants to learn, whether for work, education, or leisure, and how they might achieve those goals in the future.

3.2 Study 2 - Intra-Rural Digital Divide Comparative Study

In the second study, I will begin to examine the presence and impact of intra-rural digital divides in rural Appalachia. This study will consist of observations and interviews at each of the six public libraries in a rural county in West Virginia. My intention of looking at multiple towns within the same county is to challenge commonly used county-level rurality measures, which assign a certain level of rurality to all locations within one county.

In prior work I have conducted, I have observed stark differences between technology in libraries in different towns within this same county, which I believe point to the presence of intra-rural digital divides within this county. I anticipate that this study will shed

light on the presence and impact of intra-rural disparities in internet access, skills, and use between towns. The potential impact of intra-rural digital divides that I have referred to include differentiated access to education and jobs, which likely has implications for the socio-economic status of members of different towns, and opportunity for upward mobility.

In deploying the digital storytelling software I have described at all six libraries, I will be able to evaluate software usage metrics to evaluate engagement with technology and ability with technology at each study site. I will use general usage metrics of each piece of software (the story recording system and the story editing system) to evaluate general engagement with technology. I will also look at the data entered as part of the software usage to gain a clearer understanding of the level of skill exhibited by participants at each study site. For example, in a prior visit to one of my research sites, I found that errors in the form data entered through the story recording website seemed to indicate lower comfort with typing, with less experienced users opting not to use capitalization or punctuation.

In addition to the software usage metrics that will be captured regardless of my presence at the study sites, I will also be visiting each study site to conduct observations of technology usage at each library through classes and events. These observations will provide a more comprehensive picture of technology use that expands beyond usage of the digital literacy software. I will conduct interviews with library patrons and librarians to provide additional context for the observational data I collect. Interviews with librarians may also provide a greater picture of technology usage in the town more generally, outside of the library, as prior work on rural libraries has found that librarians are key figures in rural communities who are trusted by community members and well-informed on matters of their community [6].

4 EXPECTED OUTCOMES

4.1 Study 1 - Assess Efficacy of Culturally Informed Digital Literacy Education for Rural Appalachia

Based on results of preliminary work, which highlighted the success of digital storytelling as a method of digital literacy education, I anticipate that Study 1 will reinforce, and provide additional evidence of effective mechanisms of digital literacy education for rural Appalachian audiences. Specifically, I anticipate that findings will highlight specific technology skills that rural Appalachians want to acquire, and effective means to deliver training in those skills. I expect that desired skills will include those for professional and educational goals, and for leisure time activities. I expect that methods that align with existing tenets of rural Appalachian culture, like storytelling, will be most effective in breaking down attitude-based barriers to technology usage. I also expect to identify a balance of cultural resonance and recognition of learned skills in educational activities that will achieve ease of learning while supporting increased technology self-efficacy among participants.

4.2 Study 2 - Identify and Characterize Intra-Rural Digital Divides

In Study 2, I expect to expect to identify specific differentiated characteristics of technology interest, experience, and usage in each town. I believe that my findings will uncover less of a spectrum of more vs. less technology interest/use in each town, but a matrix of different rationales for being interested in, or avoidant of, technology usage at each study site. I also expect that some of my findings will align with more quantifiable differences between towns, such as population size and economic power. Of the six towns I will evaluate for this study, two have significant sources of either economic or governmental power. I anticipate that these two towns will have markedly higher rates of digital literacy and access than the other four towns. I also anticipate, based on findings from prior work on intra-rural divides in other sectors, that proximity to these two higher-power towns will have a positive impact on digital literacy and access. That is to say, I expect that towns located on the outskirts of these two higher-power towns will have higher rates of digital literacy and access than towns located farther away from the higher-power towns. Finally, I expect to find that residents' perceptions of resources and cultural factors in neighboring towns will impact their own self-efficacy. For example, based on preliminary work, residents of smaller towns have demonstrated a combination of resentment and admiration for the larger towns in their county. This has resulted in these residents indicating that they feel disempowered or inadequate by comparison to residents of these larger towns.

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